Bat and Ball

Starter:
https://scratch.mit.edu/projects/1270455000/editor

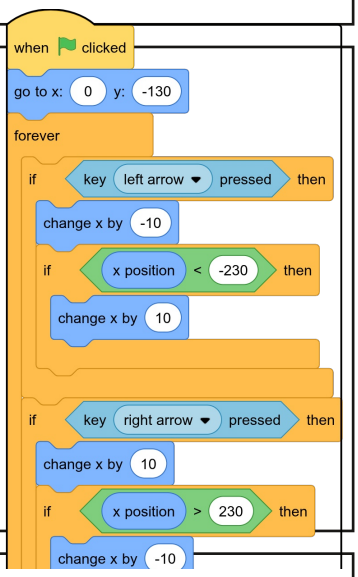
Don't forget to remix and rename.

1 The starter project has four sprites;

- **player 1** and **player 2** - these are the two bats that the players use and the code will be almost identical in each, but not quite
- the **ball** that the players try to hit - most of the code will go in this sprite
- a **countdown** to let the players get ready to play

2 The first thing to do is to let the players control their paddles. In the **player 1** sprite create a new script when the green flag is clicked to;

- put the paddle in the right place with a go to block - by setting x to 0 and y to -130 the paddle will go to the middle at the bottom of the stage
- in a forever loop detect if the left and right arrows are pressed and then move the paddle by changing its x position (left and right)
- **Note:** The picture on the right only shows you how to move left - you will need to work out how to move right!



3 The code for **player 2** is almost the same as for player 1 so you can drag the script from player 1 and drop it onto player 2 underneath the stage. Then in **player 2** you need to make a few changes;

- change **y** in the **go to** block to put the paddle at the top of the screen
- look at your keyboard and choose two different keys for left and right

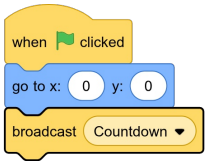
4 Next, the **countdown** sprite will let the players get ready.

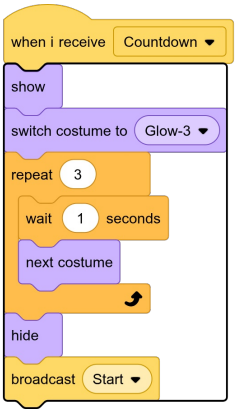
At the very start, when the green flag is clicked;

- **broadcast** the 'Countdown' message to make the countdown start

And then, in the **when I receive Countdown** block;

- show the number 3 costume
- three times, wait for a second and then show the next number costume
- after this, hide the countdown and broadcast a Start message to tell the ball to go

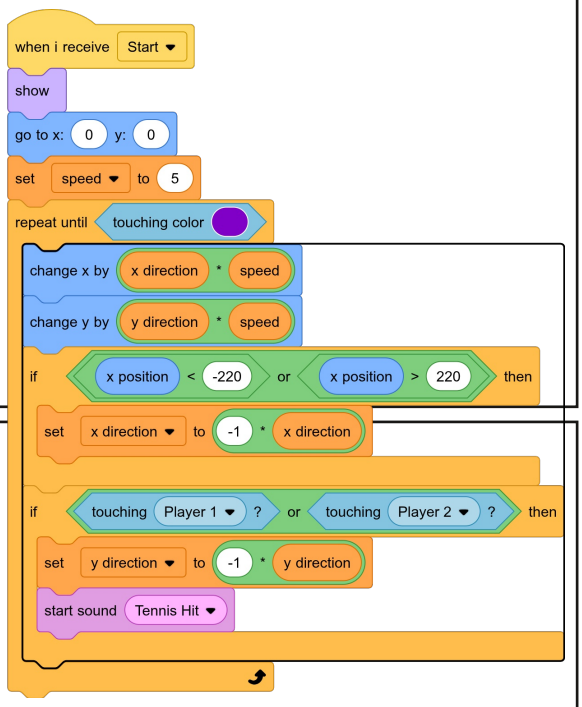




5

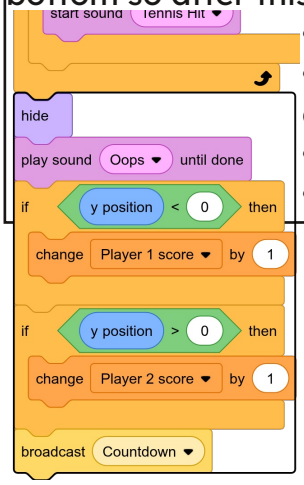
Now to get the ball moving. In the **ball** sprite;

- move the ball by changing the x and y position depending on the variables **x direction**, **y direction** and **speed**
- if the ball gets to the left or right edge then bounce - change **x direction** so it goes the opposite way
- if the ball hits one of the players' paddles then bounce - change **y direction**



6

Now to keep score and start again when one of the players misses. The while touching colour loop will stop when the ball hits the top or bottom so after this;



- hide the ball and play a sound
- if the ball is at the bottom add one to player 1's score
- if the ball is at the top add one to player 2's score
- send a message to start the countdown again

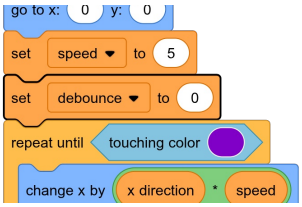
You might have wondered ...

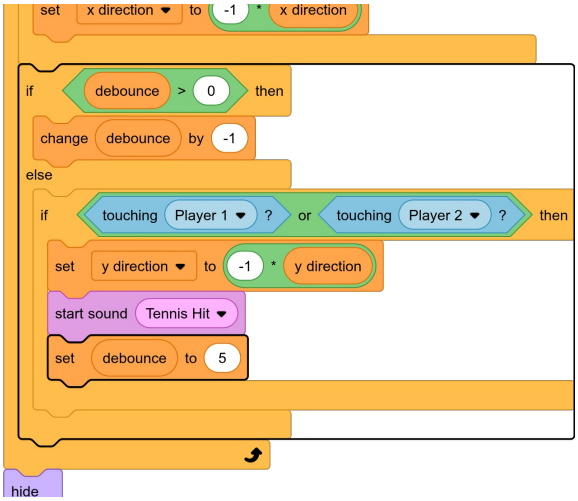
- In the countdown sprite why did we put the countdown in a **when I receive message** block and not the **when green flag clicked** block?
- It is so that the countdown could be started again in exactly the same way from here.

7

Optional (tricky!)

You might have noticed that you can "catch" the ball if you hit it with the edge of the paddle. We can fix this with **debounce.** In the script in the **ball** sprite;





- set the **debounce** variable to zero before the repeat
- set the **debounce** variable to 5 when it hits the paddle
- put the check for touching the paddle around an **if ... else** block so that if debounce is greater than 0 it will reduce it instead of checking the collision with the paddle

You might be interested ... Debounce is used in web pages to stop things happening lots of times if a button is pressed lots of times quickly.

8

What can you add to the game? Here are some ideas;

- Make the ball move faster as the game goes on.
- Stop the game when one player reaches a certain score, so the game doesn't go on forever!
- Something else from your own imagination!

© 2026 Joe Haig — Licensed CC BY-NC-SA 4.0